

# IOT-BASED FISH POND MONITORING & MANAGEMENT SYSTEM

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*Project Specification Documentation & User Manual*

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Version 1.0

Designed for Bangladesh Aquaculture

## 1. Project Overview

The IoT-Based Fish Pond Monitoring and Management System is a smart aquaculture solution designed to address the most critical challenges faced by fish farmers in Bangladesh: sudden fish death caused by deteriorating water quality, lack of continuous monitoring capability, and the absence of automated intervention mechanisms.

This system integrates a network of waterproof sensors with an ESP32 microcontroller to continuously sample water temperature, pH level, and turbidity. All readings are transmitted wirelessly over Wi-Fi to a dedicated mobile application, enabling fish pond owners to monitor their pond from anywhere in real time. Beyond monitoring, the system actively controls two actuators: an aerator (oxygen pump) and an automated fish feeder, both of which can operate on configurable schedules or be triggered manually via the app.

|                         |                                                    |
|-------------------------|----------------------------------------------------|
| Project Name            | IoT-Based Fish Pond Monitoring & Management System |
| Document Type           | Project Specification & User Manual                |
| Version                 | 1.0                                                |
| Primary Controller      | ESP32 Development Board (30-pin)                   |
| Communication           | Wi-Fi (IEEE 802.11 b/g/n)                          |
| Application Platform    | Android / iOS Mobile Application                   |
| IoT Platform            | Firebase Realtime Database / ThingSpeak            |
| Target Environment      | Fish Pond / Aquarium (Aquaculture)                 |
| Estimated Hardware Cost | 4,800 - 5,500 BDT (Base Configuration)             |
| Country / Context       | Bangladesh Aquaculture Sector                      |

## 2. Objectives

The core objectives of this project are:

1. Provide continuous, real-time monitoring of critical water quality parameters including temperature, pH, and turbidity.
2. Significantly reduce fish mortality rates by enabling rapid detection of and response to hazardous water conditions.
3. Enable remote monitoring and control of the pond ecosystem through a dedicated mobile application.
4. Implement an intelligent alert system that notifies the user via app push notification, local buzzer, and LED indicator whenever water conditions cross configured safety thresholds.
5. Automate the aerator to activate automatically when water temperature exceeds a user-defined threshold, ensuring dissolved oxygen levels remain safe.

6. Provide a flexible, configurable automated fish feeding schedule with both time-of-day scheduling and duration control.
7. Offer a cost-effective, scalable solution suitable for small-scale aquariums up to large commercial fish ponds.
8. Create a foundation for future expansion including AI-based prediction, multi-pond management, solar power integration, and SMS alerting.

## 3. System Architecture

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### 3.1 High-Level System Overview

The system is built on a three-tier architecture:

- Perception Layer: The sensor array (DS18B20, PH4502C, Turbidity Sensor) continuously samples the physical water environment and delivers analog or digital readings to the ESP32.
- Processing & Connectivity Layer: The ESP32 microcontroller processes all sensor data, applies threshold logic, controls the relay-driven actuators (aerator and feeder motor), drives the local alert system (buzzer and LED), and transmits all data to the cloud over Wi-Fi.
- Application Layer: A custom-built mobile application (Android/iOS) connects to the cloud backend, displays live sensor readings on a dashboard, allows the user to configure thresholds and schedules, and delivers push notifications when alerts are triggered.

### 3.2 Data Flow

The system follows the following data flow sequence:

9. Sensors continuously collect water quality data every configurable interval (default: every 10 seconds).
10. The ESP32 reads sensor values, applies calibration offsets, and evaluates them against configured thresholds.
11. Data is uploaded to Firebase Realtime Database over Wi-Fi.
12. The mobile application listens for real-time updates from Firebase and refreshes the dashboard instantly.
13. If a threshold is exceeded, the ESP32 triggers the local alert (buzzer + LED) and Firebase Cloud Messaging (FCM) sends a push notification to the registered mobile device.
14. User commands from the app (manual actuator control, threshold changes, schedule updates) are written to Firebase and the ESP32 listens for these changes and responds accordingly.

## 4. Hardware Components

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All components listed below have been selected for compatibility, availability in Bangladesh, and optimal price-to-performance ratio. Prices reflect current market rates from verified Bangladeshi suppliers.

| Component                | Role / Description                                                                                        | Price (BDT) | Source         |
|--------------------------|-----------------------------------------------------------------------------------------------------------|-------------|----------------|
| ESP32 Dev Board (30-pin) | Main controller; handles all sensor reading, Wi-Fi, relay control, and app communication                  | 430         | roboticsbd.com |
| DS18B20 (Waterproof)     | Measures water temperature with $\pm 0.5^{\circ}\text{C}$ accuracy; fully waterproof probe for submersion | 220         | daraz.com.bd   |
| pH Sensor PH4502C        | Measures water acidity/alkalinity (0-14 pH range); analog output with onboard signal conditioning         | 2,900       | techshopbd.com |
| Turbidity Sensor         | Detects suspended particles in water; indicates pollution levels and algae growth                         | 820         | techshopbd.com |
| Relay Module (2-channel) | Controls the aerator and feeder motor; switches high-voltage/high-current loads safely from ESP32         | 400         | roboticsbd.com |
| Buzzer + LED             | Local alert system; provides audible and visual alarm when thresholds are breached                        | 200         | roboticsbd.com |
| 18650 Battery + TP4056   | Rechargeable power supply; portable and suitable for outdoor deployment                                   | 100         | roboticsbd.com |
| Aquarium Aerator (small) | Provides oxygen to the water; controlled via relay by ESP32 automatically or manually                     | ~300        | Local market   |
| Feeder Motor (DC)        | Dispenses fish feed on schedule; driven via relay module from ESP32                                       | ~200        | Local market   |
| Supporting Components    | Breadboard, jumper wires, waterproof enclosure box, 4.7k resistor, capacitors                             | ~200        | Various        |

## 4.1 ESP32 Development Board

The ESP32 serves as the central intelligence of the entire system. It integrates a dual-core Xtensa LX6 processor running at up to 240 MHz, built-in Wi-Fi (IEEE 802.11 b/g/n), and Bluetooth, making it an ideal choice for IoT applications without requiring any additional wireless communication modules.

- Processor: Dual-core Xtensa LX6, up to 240 MHz
- Wi-Fi: IEEE 802.11 b/g/n - enables direct communication with Firebase cloud backend
- ADC: 12-bit Analog-to-Digital Converter for reading pH and turbidity sensors
- GPIO: Multiple digital I/O pins for DS18B20 (one-wire), relay control, buzzer, and LED
- Power: Operates at 3.3V logic with 5V input via USB or VIN pin

- Price: BDT 430 | Alternative: NodeMCU ESP8266 (BDT 350, less processing power)

## 4.2 DS18B20 Waterproof Temperature Sensor

The DS18B20 uses the one-wire protocol, requiring only a single data pin on the ESP32 for communication. Its fully waterproof stainless steel probe design makes it ideal for direct submersion into pond water.

- Measurement Range: -55°C to +125°C
- Accuracy:  $\pm 0.5^\circ\text{C}$  across the operating range
- Interface: Digital (One-Wire protocol)
- Power Supply: 3.0V to 5.5V (parasite or external power mode)
- Price: BDT 220

## 4.3 pH Sensor (PH 4502C Module)

The PH4502C module includes an onboard operational amplifier and signal conditioning circuit that converts the glass electrode potential difference into a readable analog voltage output (0-5V) proportional to the pH level of the water. Calibration using standard buffer solutions (pH 4.0 and pH 7.0) is required for accurate readings.

- pH Measurement Range: 0 to 14
- Operating Temperature: 0 to 60°C
- Output: Analog voltage (0-5V)
- Calibration: Required with standard pH buffer solutions
- Price: BDT 2,900 | Alternative: Gravity pH Sensor (BDT 3,750, more stable)

## 4.4 Turbidity Sensor

The turbidity sensor uses an infrared LED and phototransistor to measure the amount of light scattered by suspended particles in the water. Higher turbidity (more suspended solids or algae) results in more light scattering and a higher analog output value, which the system uses to gauge water clarity.

- Detection: Suspended particles, algae, sediment
- Output: Analog voltage
- Indicator: Indicates water pollution levels and need for water change
- Price: BDT 820 | Alternative: Higher-accuracy module at BDT 965

## 4.5 Relay Module (2-Channel)

The relay module acts as an electrically controlled switch, allowing the ESP32 (operating at 3.3V logic) to safely control devices operating at higher voltages such as the aquarium aerator. The 2-channel version simultaneously manages both the aerator and the feeder motor.

- Channels: 2 (one for aerator, one for feeder motor)
- Input Trigger: TTL level (3.3V compatible)

- Output: Controls up to 10A at 250VAC or 10A at 30VDC
- Price: BDT 400 | Alternative: Solid State Relay (faster switching, slightly higher cost)

### 4.6 Buzzer and LED

The buzzer and LED form the local alert system that provides immediate on-site warning when water quality parameters exceed configured thresholds, even without requiring mobile app connectivity.

- Buzzer Type: Mechanical or piezoelectric active buzzer
- LED: Standard 5mm LED (red recommended for alerts)
- Control: Simple digital HIGH/LOW signal from ESP32 GPIO
- Price: BDT 200 (combined)

### 4.7 Power System

For base (indoor aquarium) testing, a standard USB 5V power adapter can power the ESP32 directly. For outdoor pond deployment, the rechargeable 18650 battery system with TP4056 charging module provides portable power.

- Primary Option: 18650 Li-ion battery + TP4056 charging module (BDT 100)
- Outdoor Option: Solar panel (5W, 5V) + battery (BDT 500-700) for fully off-grid deployment
- Indoor Test: Standard 5V USB power adapter (no additional cost)

## 5. Monitored Parameters & Thresholds

The system monitors three core water quality parameters. Default thresholds are pre-configured based on standard aquaculture guidelines for Bangladesh. All thresholds are adjustable by the user through the mobile application.

| Parameter         | Normal Range       | Alert Threshold                 | Sensor Used      |
|-------------------|--------------------|---------------------------------|------------------|
| Water Temperature | 25 - 30 °C         | Below 20°C / Above 32°C         | DS18B20          |
| pH Level          | 6.5 - 8.5          | Below 6.0 / Above 9.0           | PH 4502C         |
| Turbidity         | Low (Clear Water)  | High Turbidity Reading          | Turbidity Sensor |
| Aerator Status    | ON / OFF           | Auto-ON when temp exceeds limit | Relay Module     |
| Feeder Status     | Scheduled / Manual | User-configured timer           | Motor + Relay    |

Note: The above thresholds represent general aquaculture guidelines. Different fish species may require different ranges. Users are advised to consult species-specific requirements and adjust thresholds in the mobile application accordingly.

## 6. Feature Specification

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### 6.1 Real-Time Water Quality Monitoring

The system continuously monitors water quality and displays live readings on the mobile application dashboard.

- Live display of water temperature (°C), pH value, and turbidity level
- Configurable reading interval (default: every 10 seconds)
- Historical data graph displaying trends over the past 24 hours, 7 days, and 30 days
- Color-coded status indicators: Green (normal), Yellow (caution), Red (alert)
- Timestamp of the most recent reading displayed on dashboard

### 6.2 Automated Aerator Control

The aerator is a critical device for maintaining dissolved oxygen levels in the pond. The system provides three modes of aerator control:

- Automatic Threshold Mode: The ESP32 automatically activates the aerator when water temperature exceeds the configured upper threshold (default: 32°C). The aerator runs for a user-configured duration (default: 15 minutes) and then stops. It re-evaluates the condition on the next sensor reading cycle.
- Scheduled Timer Mode: The user configures up to 3 time-of-day schedules (e.g., 06:00 AM, 12:00 PM, 06:00 PM) through the app. On each schedule, the aerator runs for a configured number of seconds.
- Manual Mode: The user can turn the aerator ON for any configured duration (in seconds) at any time via the app.
- The threshold trigger duration, scheduled times, and manual duration are all independently configurable.

### 6.3 Automated Fish Feeder Control

The fish feeder motor dispenses feed into the pond. The system provides:

- Scheduled Timer Mode: The user configures the exact time(s) of day the feeder should activate (e.g., 07:00 AM and 06:00 PM). Up to 5 daily schedules are supported.
- Duration Control: For each scheduled activation, the user configures how many seconds the feeder motor should run. This controls how much feed is dispensed.
- Manual Mode: The user can trigger the feeder instantly from the app for a configured number of seconds.
- Status Feedback: The app shows whether the feeder is currently running and the time of the last feeding event.

### 6.4 Alert & Notification System

The alert system operates at two levels simultaneously:

- **Local Alert (on-site):** The buzzer sounds and the LED flashes whenever any monitored parameter (temperature, pH, or turbidity) crosses its configured threshold. This occurs even without internet connectivity.
- **Push Notification:** The app sends a push notification to the registered mobile device via Firebase Cloud Messaging (FCM). The notification message includes the parameter name, current value, and threshold that was exceeded (e.g., 'ALERT: Water Temperature is 34°C, exceeding the 32°C threshold').
- **Notification Cooldown:** To prevent notification flooding, a minimum 10-minute cooldown is enforced between repeated alerts for the same parameter.
- **Alert Log:** All alerts are logged with timestamp and parameter value in the app for review.

## 6.5 User Configurable Settings

Through the mobile application, users can configure the following settings without any technical knowledge:

- Temperature alert threshold (upper and lower limits)
- pH alert threshold (upper and lower limits)
- Turbidity alert threshold
- Aerator auto-trigger temperature threshold
- Aerator auto-run duration (in seconds/minutes)
- Aerator daily schedule (up to 3 time slots)
- Feeder daily schedule (up to 5 time slots)
- Feeder run duration per schedule (in seconds)
- Manual aerator run duration (in seconds)
- Manual feeder run duration (in seconds)
- Sensor reading interval

## 6.6 Offline Resilience

The system is designed to continue functioning even when internet connectivity is temporarily unavailable:

- The ESP32 stores the last known configuration (thresholds, schedules) in non-volatile memory (NVM flash). Schedules and threshold-based automation continue to operate offline.
- Local alerts (buzzer and LED) function entirely without internet connectivity.
- Data is buffered locally and synced to Firebase once connectivity is restored.

# 7. Mobile Application Overview

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## 7.1 Application Screens

- **Home Dashboard:** Real-time display of all three sensor readings with color-coded status. Shows aerator and feeder status. Quick-access manual control buttons.



- Historical Data Screen: Line graphs for temperature, pH, and turbidity over selectable time ranges (24h, 7 days, 30 days).
- Aerator Control Screen: View current status. Toggle manual on/off with duration input. View/edit scheduled times and durations. Configure auto-trigger threshold.
- Feeder Control Screen: View last feeding time. Trigger manual feed with duration input. View/edit up to 5 daily scheduled feeding times and durations.
- Alert Settings Screen: Configure threshold values for all three parameters. View alert history log.
- Notification History Screen: List of all past push notifications with timestamps.
- Device Settings Screen: Wi-Fi configuration, sensor calibration offset input, reading interval setting.

## 7.2 Application Platform

- Primary Platform: Android (Android 8.0 and above)
- Future Support: iOS (via cross-platform Flutter framework)
- Connectivity: Firebase Realtime Database for live data sync, Firebase Cloud Messaging for push notifications

## 8. Total Cost Estimation

| Item                                           | Unit Price (BDT)     | Notes                     |
|------------------------------------------------|----------------------|---------------------------|
| ESP32 Development Board (30-pin)               | 430                  | Main controller           |
| DS18B20 Waterproof Temperature Sensor          | 220                  | Water temperature         |
| pH Sensor Module (PH4502C)                     | 2,900                | pH measurement            |
| Turbidity Sensor Module                        | 820                  | Water clarity             |
| 2-Channel Relay Module                         | 400                  | Controls aerator & feeder |
| Buzzer + LED Indicator                         | 200                  | Local alert               |
| 18650 Battery + TP4056 Charger                 | 100                  | Power supply              |
| Aquarium Aerator (small)                       | ~300                 | Oxygen pump               |
| DC Feeder Motor                                | ~200                 | Feed dispenser            |
| Supporting (wires, breadboard, box, resistors) | ~200                 | Miscellaneous             |
| <b>TOTAL ESTIMATED COST</b>                    | <b>5,570 - 6,000</b> | Base configuration        |

Note: Software development (mobile app) uses free-tier Firebase services. No licensing cost is required for the base configuration. Prices may vary by supplier and availability.

## 9. Future Expansion Roadmap

The system has been architecturally designed for scalability. The following features are planned for future versions:

- **AI-Based Prediction:** Machine learning model to predict water quality deterioration based on historical trends, enabling preventive action before thresholds are breached.
- **Multi-Pond Management:** Support for multiple ESP32 nodes (ponds) within a single mobile application with a centralized dashboard.
- **SMS Alert System:** Integration with a GSM SIM800L module (BDT 350-450) or an SMS API to send alerts even when the user has no internet connection.
- **Solar Power Integration:** Solar panel (BDT 500-700) and rechargeable battery system for fully autonomous outdoor deployment.
- **Water Quality Report Export:** Weekly/monthly PDF report generation from historical data, emailed to the pond owner.
- **Fish Health Monitoring:** Dissolved oxygen sensor and ammonia sensor additions to the sensor array.
- **Commercial Scale:** LoRa long-range communication (BDT 600-900) for large ponds where Wi-Fi does not reach.

## 10. User Manual

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### 10.1 Getting Started

Before using the system for the first time, complete the following setup steps:

1. Download and install the Fish Pond Manager application from the provided APK file (Android) or app store link.
2. Power on the ESP32 unit by connecting it to a 5V USB power adapter or the battery system.
3. Open the app and navigate to Device Settings. Enter your Wi-Fi network name (SSID) and password. The ESP32 will connect automatically.
4. Once connected, the app dashboard will begin displaying live sensor readings within 30 seconds.
5. Calibrate the pH sensor: navigate to Device Settings > pH Calibration. Follow the on-screen instructions using pH 7.0 and pH 4.0 buffer solutions.
6. Configure your alert thresholds by navigating to Alert Settings. Adjust temperature, pH, and turbidity thresholds to match your fish species requirements.
7. Set up feeding schedules by navigating to Feeder Control > Schedule. Add feeding times and durations.

### 10.2 Reading the Dashboard

- **Temperature Card:** Displays current water temperature in °C. Background is green when normal, yellow when approaching threshold, red when threshold is exceeded.
- **pH Card:** Displays current pH value. Green for 6.5-8.5, yellow for 6.0-6.5 or 8.5-9.0, red outside this range.
- **Turbidity Card:** Displays turbidity reading. Green for low (clear water), yellow for moderate, red for high turbidity.
- **Aerator Status:** Shows ON/OFF status and the remaining duration if running on a timer.

- Feeder Status: Shows the last feeding time and the next scheduled feeding time.

### 10.3 Manual Control

To manually run the aerator:

8. Navigate to Aerator Control on the app.
9. Set the desired run duration in seconds using the duration input field.
10. Tap 'Run Now'. The aerator will run for the specified duration and stop automatically.

To manually trigger the feeder:

11. Navigate to Feeder Control on the app.
12. Set the desired run duration in seconds.
13. Tap 'Feed Now'. The feeder motor will run for the specified duration.

### 10.4 Configuring Schedules

To set up aerator or feeder schedules:

14. Navigate to the respective control screen (Aerator Control or Feeder Control).
15. Tap 'Add Schedule'.
16. Select the time of day (hour and minute) for the scheduled activation.
17. Enter the run duration in seconds.
18. Tap 'Save'. The schedule will be shown in the list and synced to the ESP32 automatically.
19. To delete a schedule, swipe left on the schedule item and tap 'Delete'.

### 10.5 Understanding Alerts

When an alert is triggered:

- A push notification appears on your phone with the alert details.
- The local buzzer sounds and the LED flashes at the pond location.
- The affected sensor card on the dashboard turns red.
- The alert is logged in the Notification History screen.

To clear the local buzzer/LED alarm manually, tap 'Acknowledge Alert' in the app. This silences the local alarm until the condition returns to normal range.

### 10.6 Troubleshooting

- No sensor readings on dashboard: Check that the ESP32 is powered and the Wi-Fi indicator LED is solid (not blinking). Verify Wi-Fi credentials in Device Settings.
- pH readings seem inaccurate: Recalibrate the pH sensor. Go to Device Settings > pH Calibration and follow the procedure with fresh buffer solutions.
- Aerator not responding to commands: Check the relay module wiring and ensure the relay power supply is adequate. Verify the aerator is plugged into the relay output correctly.
- Feeder not dispensing at scheduled time: Verify the ESP32's real-time clock is synchronized. Open the app and check Device Settings > Sync Time. Ensure the ESP32 has internet access.

- App not receiving notifications: Check that notification permissions are enabled for the app in your phone's Settings. Verify FCM token refresh in Device Settings.

## 11. Safety & Compliance

- The ESP32 unit must be housed in a waterproof enclosure (IP65 or higher rating) when deployed near water.
- All 240V AC electrical connections (if using an AC aerator) must be performed by a qualified electrician or using a UL-rated power socket. The relay module handles the switching, but AC connections are the user's responsibility.
- Do not submerge the ESP32 unit, relay module, or any electronics. Only the waterproof sensor probes are rated for water contact.
- Ensure the power supply is protected from moisture and rain in outdoor deployments.
- Disconnect power before making any changes to wiring connections.

## 12. Technical Specifications Summary

| Specification           | Value                                                     |
|-------------------------|-----------------------------------------------------------|
| Microcontroller         | ESP32 (Dual-core Xtensa LX6, 240 MHz, 4MB Flash)          |
| Connectivity            | Wi-Fi IEEE 802.11 b/g/n (2.4 GHz)                         |
| Temperature Range       | -55°C to +125°C (DS18B20)                                 |
| Temperature Accuracy    | ±0.5°C                                                    |
| pH Range                | 0 to 14 pH (PH4502C)                                      |
| ADC Resolution          | 12-bit (0-4095 raw value)                                 |
| Relay Channels          | 2 (Aerator + Feeder)                                      |
| Max Relay Load          | 10A / 250VAC per channel                                  |
| Alert Outputs           | Buzzer (local) + LED (local) + Push Notification (mobile) |
| Cloud Backend           | Firebase Realtime Database + Firebase Cloud Messaging     |
| App Platform            | Android (Flutter framework, iOS future)                   |
| Sensor Reading Interval | 10 seconds (configurable, min: 5s, max: 60s)              |
| Data Retention (Cloud)  | 30 days (configurable, Firebase free tier)                |
| Max Feeder Schedules    | 5 per day                                                 |
| Max Aerator Schedules   | 3 per day                                                 |
| Power Supply            | 5V USB (indoor) / 18650 battery / Solar (outdoor)         |
| Operating Temperature   | 0°C to 50°C (electronics enclosure)                       |

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*End of Project Specification & User Manual*

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